

AAYAM BAJAJ

✉ aayambajaj4@gmail.com ☎ 9516644441 🌐 GitHub 🔗 Portfolio 🌐 LinkedIn 📍 Indore / Mumbai

I'm an IT engineer passionate about building impactful tech solutions across real-time systems, machine learning, and web development. Experienced in leading projects and collaborating in team environments, with strong skills in problem-solving, communication, and organization. Always eager to learn, take initiative, and apply technology to solve real-world challenges.

WORK EXPERIENCE

Java Development Internship — ShadowFox Technologies LLP, Chennai (June 2025) | **Web Development Internship** — RWS Moravia India Pvt. Ltd. (U.K.) (July 2024) | **Cyber Crime Investigation, Prosecution, Forensics & Cyber Security** — Madhya Pradesh State Cyber Cell, Indore (June 2024) | **Training - Cybersecurity** — AcmeGrade Pvt. Ltd., Bengaluru (Oct-2023)

EDUCATION

Bachelor of Engineering in Information Technology 2022 – 2026 | Navi Mumbai
Bharati Vidyapeeth College of Engineering, Navi Mumbai
CGPA: 8.37

XII (ISC), Laurels School International, Indore 2021 – 2022 | Indore
Percentage: 93%

X (ICSE), Laurels School International, Indore 2019 – 2020 | Indore
Percentage: 94%

PROJECTS

Smart Planner – Accessible Route Optimization System 🔗

- Developed an accessibility-focused navigation system using Flask and OpenStreetMap with a modified A* algorithm for optimized routing.
- Integrated ML-based obstacle prediction and dynamic cost modeling based on slope, surface, and weather.

Agentic City Planner – Simulation Engine 🔗

- Developed a FastAPI-based simulation engine to model traffic, cost, and pollution for urban planning decisions.
- Built modular APIs with Pandas and NumPy for real-time analysis and integration with an AI planning system.

Crowd Analysis using Edge Computing 🔗

- Built a real-time crowd detection system using TensorFlow Lite on Raspberry Pi with YOLOv8 validation.
- Developed a Tkinter GUI and Flask dashboard, achieving sub-500ms latency for live monitoring.

PUBLICATIONS

Aayam Bajaj, et al., **Dynamic Multi-Objective Cost Function for Personalized and Accessible Urban Pathfinding** 🔗, IEEE Xplore, 2026

SKILLS

Languages — Java, Python | **Web** — HTML, CSS, JavaScript, AngularJS, Flask | **Databases** — MySQL, MongoDB |
Tools & Technologies — Git, GitHub, FastAPI, Raspberry Pi | **Core Concepts** — Data Structures & Algorithms, OOP, DBMS |
Soft Skills — Public-Speaking, Event Management, Group Leadership

ACHIEVEMENTS

Chief Editor, Magazine (College) | Winner, College-level Technical Quiz | Joint General Secretary, Student Council (College) |
Leader, Literature Club (College)

CERTIFICATIONS

Data Structures and Algorithms using Java 🔗, NPTEL

Data Analytics with Python 🔗, NPTEL

Java (Basic), SQL (Basic), Python (Basic) 🔗, HackerRank